p-hacking HW

Homeworks are now for 2 classes. They will take twice as long and count twice as much. Same rules apply.

- In a markdown cell write down the AI model you use if needed
- This Homework is following the notes. Use the notes!!!!
- You will see each question and my answers below. Your HW should look very similar but with code and some more epxlanation.
- If you get stuck try the notes. Try and find the cell in the notes where we did something similar.
- If you are truly stuck then come to office hours, talk to a friend, use google or AI for help. If you use help on a question write down what you asked and what you learned
- I am not sure how long this HW will take.

AI Model:

Overview

- I did field work in Bangladesh with a colleague from Britain (Edwin Gnanaprakasam). Edwin then went and did illumina sequencing to look at microbial species diversity in Bangladesh sediment. He did 16s rRNA analysis of the DNA and found percent species abundance. I then did analysis of the water and sediment to get the arsenic content. Edwin wanted to know what bacterial species from this large dataset he should further investigate. Now we want to look at bacterical species and see if there are correlations with the the different arsenic species. He is asking if you can give him a nice notebook and figure. Here is a link to the paper we published.
- Your mission if you choose..... Is to p-hack this data. We are going to split of the class to look at different sites and different data.
- If you do this homework and structure it well you should have a notebook that can work for almost any data file in the future when examining correlations

Q1

- Read in the Fall2026ClassList.xlsx with sheet 'class list' and set the index_col to uni.
- You will need the updated file.
- Your uni should be in the class list.
- Use your pandas nomenclature to show the values associated with your uni.
- This will give you the data you need to analyze on this and future homeworks.
- You can look in the excel file to see how this is set up. But you need to do the pandas call to see it also.

Out[62]:
```
Name              Mailloux, Brian
WellData                    Drink
SiteName                    SiteB
ArsenicSpecies              As2S3
NYCTree                    status
Core#                  18UWHC004
Poster Group                   4
Name: bjm2103, dtype: object
```

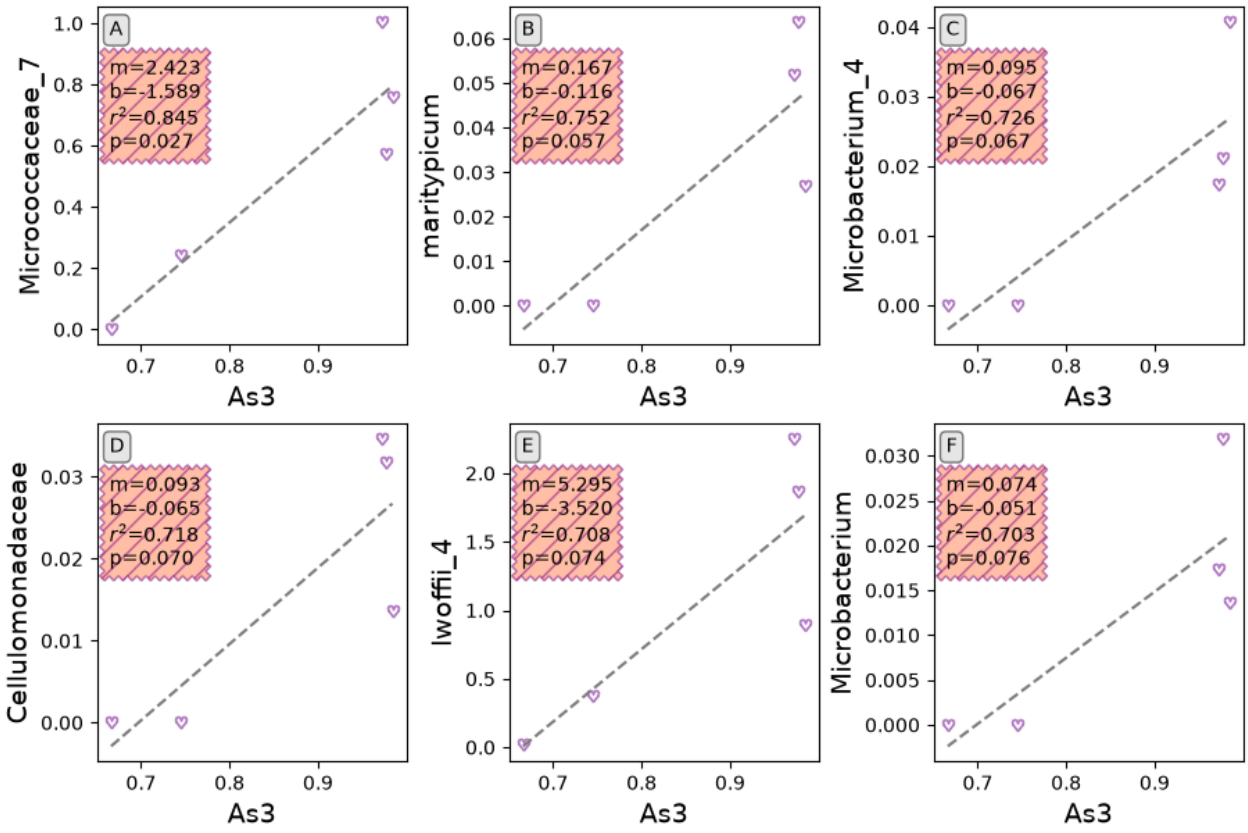
Q2 - All correlations in a PDF file

- read in the Species-Arsenic-Data-Cleaned.xlsx file.
- read in the sheet from your site
- use the arsenic species from Class list to correlate to the bacterial speices
- drop the other columns from the names of columns you don't want to correlate to
- You just want to look at bacterial species.
- Make a pdf file that contains a plot of As versus every other parameter.
- Make sure it is labeled nicely and in one pdf file.
- Arsenic species should be on the x-axis and the other parameters should be on the y-axis.
- each figure needs a best fit line and the results
- Each parameter should be on one page.
- make sure your axes go in the correct direction. We are not plotting depth anymore
- your graphs should be square.
- Choose a color and symbol you like for your plots
- You will need to hand in your pdf file of results.
- your plot will not show up in the notebook.
- you need to save all your results data to a new dataframe

```
Successfully exported 561 pages to Results/SiteB_As3.pdf!
Successfully exported 561 data to Results/SiteB_As3.xlsx!
```

Q2 Subplots of top fits!

- Make a plot with subplots that is your As sepecies versus the 6 bacterial species with the best fits
- Save this graph to a jpeg and turn that in .
- Make sure your graphs look good and have the best fit data
- They should have letters denoting them like in the packet
- In a markdwon cell note how many figures are statisitcall significant at what level. you might need to look in your pdf file.
- Make a nice table in markdown.



How Many species are signicant?

- The table shows the number of species significant at each level.

Significance Level	Number of Species
p<0.05	????
p<0.01	%%%%
p<0.001	****
Sum	#####

Q3

- Make a Seaborn Heatmap to see if the top 20 species that are correlated with arsenic
- This is a three part question
- Part1: show your df_r2 dataframe
- Part2: show the top 20 species
- Part3: make the heatmap
- Part4: Comment in 1 or 2 sentences what the results mean for the correlations with arsenic and the other microbial species.

Q3-1

Out[47]:

	Xcolumn	Ycolumn	r2	rvalue	pvalue	slope	intercept	stderr	n_sam
176	As2S3	Syntrophobacteraceae_2	0.998701	0.999350	0.000020	0.262488	-0.000701	0.005466	
438	As2S3	BSV26	0.986690	0.993323	0.000654	0.388789	-0.003366	0.026071	
543	As2S3	Sphingomonadaceae_6	0.986690	0.993323	0.000654	0.194395	-0.001683	0.013035	
151	As2S3	Aurantimonadaceae	0.982470	0.991196	0.000990	0.502491	0.003866	0.038752	
346	As2S3	Ellin6067	0.972892	0.986353	0.001910	0.434397	0.002884	0.041864	
...	...	...	...	...	...	...	...	...	
46	As2S3	Acinetobacter	0.000037	-0.006046	0.992301	-0.010687	0.263060	1.020449	
266	As2S3	Sphingomonas_3	0.000014	0.003718	0.995266	0.000636	0.016787	0.098750	
36	As2S3	Arthrobacter	NaN	NaN	NaN	0.000000	0.000000	NaN	
313	As2S3	Spirosoma	NaN	NaN	NaN	0.000000	0.000000	NaN	
432	As2S3	Sanguibacteraceae	NaN	NaN	NaN	0.000000	0.000000	NaN	

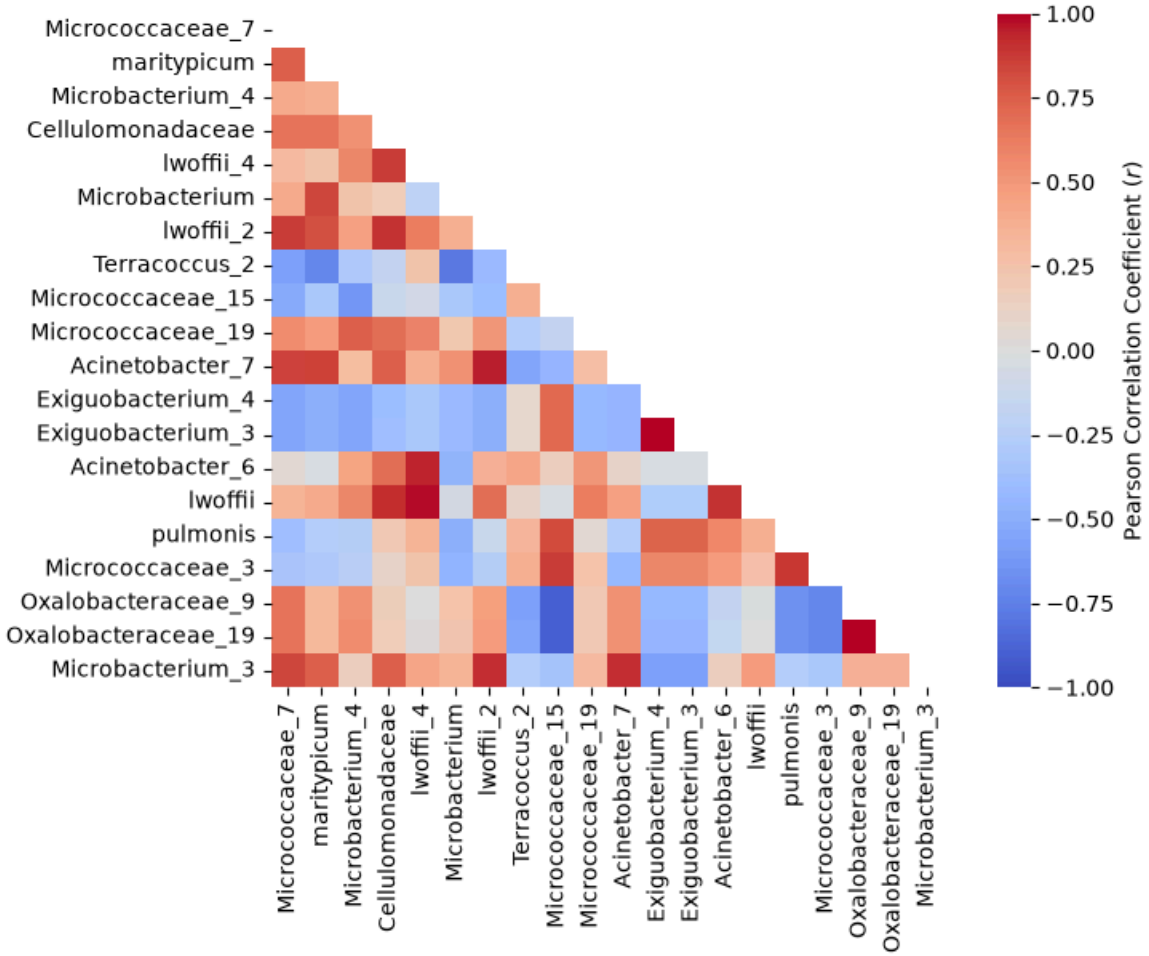
561 rows × 9 columns

Q3-2

Out[56]:

```
<StringArray>
[   'Micrococcaceae_7',         'maritypicum',    'Microbacterium_4',
   'Cellulomonadaceae',           'lwoffii_4',      'Microbacterium',
           'lwoffii_2',       'Terracoccus_2',   'Micrococcaceae_15',
   'Micrococcaceae_19',     'Acinetobacter_7',   'Exiguobacterium_4',
   'Exiguobacterium_3',     'Acinetobacter_6',             'lwoffii',
            'pulmonis',    'Micrococcaceae_3',  'Oxalobacteraceae_9',
 'Oxalobacteraceae_19',    'Microbacterium_3']
Length: 20, dtype: str
```

Q3-3



Q3-4

The results indicate.....